Me[DL1]mbrane Air Separation

CHE 381 - Chemical Engineering Lab I
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Introduction

Purpose of the lab:

- To determine the effects of pressure on air separation in series and parallel connection
- Done by calcu[DL2] the transport coefficients and recoveries [DL4]
- Separation de[DL3]s on relative rates of mass transfer, not phase equilibrium relations or pores [DL5]

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[DL10] [DL11]

Theory

- Pressure gradient allows air to move through separator modules
- Transport occurs using solution-diffusion mechanism

The diffusion flux is determined by Fick's Law: [DL6] [DL7]

$$J_A = D_A\left(\frac{C_{A1} - C_{A2}}{z}\right) \cdots (1)$$

Where,

J_A = flux of component A in $\frac{mol}{m^2 s}$
D_A = diffusivity of component A in $\frac{m^2}{s}$
z = thickness of membrane in meters
C_{A1} = concentration of component A inside membrane wall on feed in $\frac{mol}{m^3}$
C_{A2} = conc. of component A outside membrane wall on permeate side in $\frac{mol}{m^3}$

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Theory

We need to apply Henry's Law:

$$C_A = p_A S_A \cdots (2)$$

Where,

C_A = concentration of component A inside membrane wall in $\frac{mol}{m^3 Pa}$
p_A = partial pressure of component A in the gas phase in Pa
S_A = solubility constant for component A in the membrane in $\frac{mol}{m^3 Pa}$

We also incorporate the equation for the permeability coefficient:

$$\wp_A = D_A S_A \cdots (3)$$

D_A = diffusivity of component A in $\frac{m^2}{s}$
S_A = solubility constant for component i in the membrane in $\frac{mol}{m^3 Pa}$
$\wp_A$ = permeability coefficient, often expressed in Barrer, where; 1 Barrer $= 10^{-10} cm^3 (STP) cm/cm^2 s \cdot cmHg$

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Slide 1

DL1 94/100
Donglin Li, 10/8/2019

Slide 2

DL2 Use bigger fonts.
Donglin Li, 10/8/2019

DL3 -1
Donglin Li, 10/8/2019

DL4 Rephrase these sentences. That's not purposes.
Donglin Li, 10/8/2019

DL5 -1
Donglin Li, 10/8/2019

Slide 3

DL6 Use Equation Editor.
Donglin Li, 10/8/2019

DL7 -1
Donglin Li, 10/8/2019

DL10 Where's the equation for mass transfer coefficients?
Donglin Li, 10/8/2019

DL11 -1
Donglin Li, 10/8/2019

Theory

-Alpha prime: Separation factor based on the retentate composition

-Alpha double prime: Separation factor based on feed composition

$$\alpha'_{AB} = \frac{\left(\frac{x_{Ap}}{x_{Bp}}\right)}{\left(\frac{x_{Ar}}{x_{Br}}\right)}$$

$$\alpha''_{AB} = \frac{\left(\frac{x_{Ap}}{x_{Bp}}\right)}{\left(\frac{x_{Af}}{x_{Bf}}\right)}$$

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Theory

We combine those 3 equations to get new diffusion equation we can solve:

$$J_A = Q_A(P_{A1} - P_{A2}) \text{ ---(4)}$$

Where Q_A is the permeability, P_{A1} is the partial pressure of component A on the feedside, and P_{A2} is the particle pressure of component A on the permeate side.

We can then calculate the recovery of oxygen and nitrogen:

$$O_2 Recovery = \left(\frac{V_p x_{O_2,p}}{V_f x_{O_2,f}}\right) \text{ ---(5)}$$

DL8
DL9

$$N_2 Recovery = \left(\frac{V_p x_{N_2,p}}{V_f x_{N_2,f}}\right) \text{ ---(6)}$$

Where:

V_p = volumetric flowrate of permeate in $\frac{m^3}{s}$

V_f = volumetric flowrate of feed

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Apparatus

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Material and Supplies

-Oxygen Sensor and batteries

-Compressed air from cylinder

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DL8 Mention the meanings of every term in your equation.
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DL9 -1
Donglin Li, 10/8/2019

Procedure

1.Twist the valve on the pressure regulator to get it open

2.Turn on the battery powered oxygen analyzers (wait 10 minutes)

3.Calibrate oxygen powered analyzers using given procedure

4.Set the system to the parallel configuration(Valve A left, Valve B down to vent, Valve C down, Valve D down to vent) DL13

5.Adjust the pressure using the valve and the flow rate using the flowrate valve

6.Run 13 trials at varying pressure and flow rates

7.Repeat procedure with the series configuration(Valve A right, Valve B down to vent, Valve C up and Valve D down to vent

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Experimental Challenges

- Allow time for the oxygen meter to adjust the oxygen concentration should be consistently increasing or decreasing with pressure
- The apparatus valve system is sensitive and a dedicated team member should be decided to moderate pressure to system
- Be sure to properly align valves for parallel and series configurations

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Safety

- The apparatus contains multiple electrical components be sure no exposed wires exist and minimize water contact.
- The air tank is pressurized near 2500 psig therefore be conscientious that the apparatus does not receive pressure greater than 200 psig
- Always wear eye protection incase high velocity shrapnel exits the apparatus

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DL12 Typesetting is not good. Too close to the bottom.
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DL13 -1
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